

Emotional Assistance*

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Abstract—This paper presents a novel therapeutic conversation agent that integrates multimodal emotion recognition technologies with standardized mental health assessments to deliver personalized therapeutic interactions. The system combines facial emotion recognition through computer vision, speech emotion recognition through audio analysis, and standardized psychological questionnaires to create a comprehensive understanding of a user’s emotional and mental state. This information goes to the logic agent to make a system prompt appropriate for a therapeutic dialogue to a large language model. Initial evaluations show promising results in accurately detecting emotions and providing emotional support. This paper details the architecture, implementation and evaluation of this integrated system. This system has potential applications in supplementary mental health support, remote patient monitoring, and as an accessible first-line resource in regions with limited access to mental health professionals. The multimodal approach enables more nuanced and personalized support than text-only systems, potentially increasing user trust and adherence.

I. INTRODUCTION

Mental health conditions affect a significant portion of the global population. In 2019, 970 million people globally were living with a mental disorder, with anxiety and depression the most common. Besides, People with severe mental health conditions die 10 to 20 years earlier than the general population [1]. Despite the prevalence and impact of these conditions, access to mental health services remains limited due to factors including stigma, cost, and availability of qualified providers. Digital mental health interventions offer promising alternatives to traditional care models, potentially increasing accessibility and reducing barriers to treatment. Also, randomized controlled trials demonstrate that fully generative-AI therapy chatbots can significantly reduce clinical-level symptoms of anxiety and depression, with one RCT showing effect sizes comparable to low-intensity psychological interventions [2]. Emotion recognition plays a critical role in understanding user needs and enhancing communication. Facial Emotion Recognition (FER) refers to the process of identifying and categorizing human emotions based on facial expressions. By analyzing facial features and patterns, machines can learn about a person’s emotional state. This subfield of facial recognition is highly interdisciplinary, drawing on insights from computer vision, machine learning, and psychology [3]. Speech Emotion Recognition is a sub-discipline of affective computing that involves recognizing the emotional aspects of speech irrespective of the semantic content. A typical SER system can be considered as a collection of methodologies that isolate, extract and classify speech signals to detect

emotions embedded in them [4]. This project is about a system that integrates multiple emotion recognition modalities—facial expression analysis and speech emotion recognition—with standardized mental health assessments to inform a therapeutic conversation agent. The system aims to create a more empathetic and effective digital mental health intervention by combining these various inputs to gain a comprehensive understanding of the user’s emotional and mental state.

II. RELATED WORKS

A. Facial Emotion Recognition

Facial emotion recognition has been extensively studied in computer vision, with approaches ranging from geometric feature-based methods to deep learning models. Convolutional Neural Networks (CNNs) have demonstrated particularly strong performance on benchmark datasets such as FER2013 [5]. These models typically classify facial expressions into a discrete set of basic emotions, such as those proposed by Ekman [6]: anger, disgust, fear, happiness, sadness, and surprise, with neutrality often included as an additional category.

B. Speech Emotion Recognition

Early Speech Emotion Recognizer relied on spectral and prosodic features-MFCCs, ZCR, pitch, and energy—fed into classifiers like SVMs or GMMs. Recent trends emphasize deep representation learning, automatically extracting hierarchical features from raw audio or spectrograms, significantly improving robustness and generalization. Lightweight ensemble models combining CNNs and Bi-LSTMs have demonstrated competitive accuracy on benchmarks such as RAVDESS and TESS, especially when augmented with noise addition and spectrogram shifting. Comprehensive reviews also highlight challenges in class imbalance, cross-corpus generalization, and real-time deployment for low-resource devices[7].

C. Mental Health Questionnaires

Standardized mental health questionnaires are widely used in clinical settings for screening, diagnosis, and monitoring of mental health conditions. Common instruments include the Patient Health Questionnaire-9 (PHQ-9) [8] for depression, the Generalized Anxiety Disorder-7 (GAD-7) [9] for anxiety, the Perceived Stress Scale (PSS) for stress [10], and the WHO-5 Well-Being Index for overall psychological well-being [1]. These tools have been validated in numerous studies and provide valuable quantitative measures of mental health status.

D. LLM-based chatbot

Large language models (LLMs) have been rapidly adopted to power mental health chatbots, with over 50 systems identified—22 of which are explicitly LLM-based—addressing general distress, depression, anxiety, stress, and suicidality through conversational interfaces [11]. Randomized controlled trials (RCTs) demonstrate that expert-fine-tuned LLM chatbots can match low-intensity psychological interventions: an RCT of the “Therabot” system showed clinically significant reductions in anxiety and depressive symptoms over eight weeks (Cohen’s $d \approx 0.5$ – 0.6) [2], and a follow-up trial reported a mean symptom reduction for depression of 51% and 31% for generalized anxiety disorder after two months of use. A separate pilot study found that nearly 80% of participants reported ChatGPT helped them manage stress through mindfulness and relaxation exercises [12].

III. FACIAL EMOTION RECOGNITION

A. VGG-like with FER2013

1) *Introduction:* The goal is to develop a deep learning model that can accurately classify facial expressions into one of six categories: Angry, Fear, Happy, Sad, Surprise, and Neutral. I try a CNN model with a VGG-like structure to train the model with the FER2013 dataset.

2) *Data:* The FER-2013 dataset consists of 48x48 pixel grayscale images of faces that have been automatically registered to be centred and occupy a similar amount of space in each image. The dataset contains 24,400 images, with 22,968 examples in the training set and 1,432 examples in the public test set.

3) *Preprocessing:* I used data generators and data augmentation to train the deep learning model:

- Data generators create batches of images on-the-fly during training, making it possible to handle this large and complex dataset efficiently. Since loading all images into memory at once would be impractical, data generators allow the model to be trained on the entire dataset without exceeding memory limits.
- Data augmentation techniques to the training images, such as rotation, shifting, and flipping for increasing the size and diversity of the training set, which can improve the performance of the deep learning model. By introducing variability into the training process, data augmentation can also help to prevent overfitting, which occurs when the model becomes too closely tailored to the training set and performs poorly on new, unseen data.

4) *Training:* To develop the CNN, at first I defined the input layer and the number of filters in the first convolutional layer. Then, I added additional convolutional layers with increasing numbers of filters, followed by max-pooling layers to reduce the spatial dimensions of the feature maps. After the convolutional layers, I added fully connected layers with ReLU activation to classify the emotions.[13]

I experimented with different numbers of convolutional layers, filter sizes, and fully connected layers to optimize the

model performance. Additionally, I used techniques such as dropout and batch normalization to prevent overfitting and improve the generalization capability of the model.

The structure that I finally chose is:

TABLE I
SUMMARY OF THE CNN ARCHITECTURE

Layer	Filters	Kernel Size	Activation
Input	-	$48 \times 48 \times 1$	-
Conv2D + BatchNorm	32	3×3	ReLU
Conv2D + BatchNorm	64	3×3	ReLU
MaxPooling2D + Dropout	-	2×2	-
Conv2D + BatchNorm	128	3×3	ReLU
Conv2D + BatchNorm	128	3×3	ReLU
MaxPooling2D + Dropout	-	2×2	-
Conv2D + BatchNorm	256	3×3	ReLU
Conv2D + BatchNorm	256	3×3	ReLU
MaxPooling2D + Dropout	-	2×2	-
Flatten	-	-	-
Dense + BatchNorm	256	-	ReLU
Dropout	-	-	0.5
Dense (Output)	7	-	Softmax

5) *Evaluation:* The training and validation loss function decreased over epochs with a final loss in the epoch 50 of loss: 0.9036 that it typically means the model is not performing optimally.

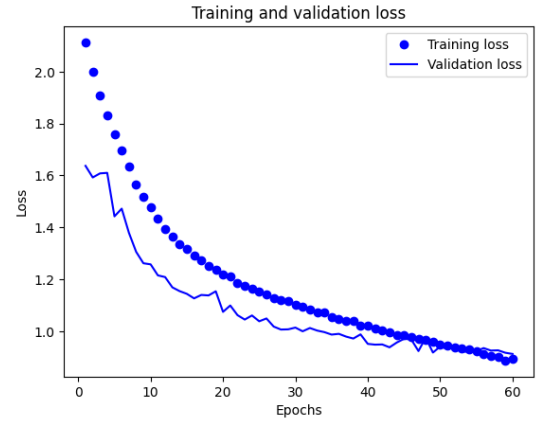


Fig. 1. Training and validation loss graph.

The training and validation accuracy increased over epochs with a final accuracy of 0.6631 that is not enough.

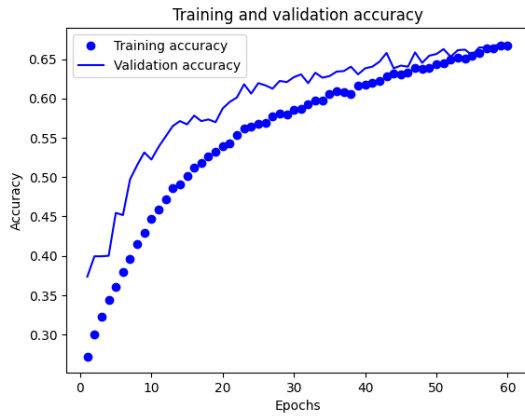


Fig. 2. Training and validation accuracy graph.

In the confusion matrix there are a lot of false positives and false negatives showing not enough accuracy. Besides, there are 0 disgust labeled images in the validation, this shows an imbalance in the data.

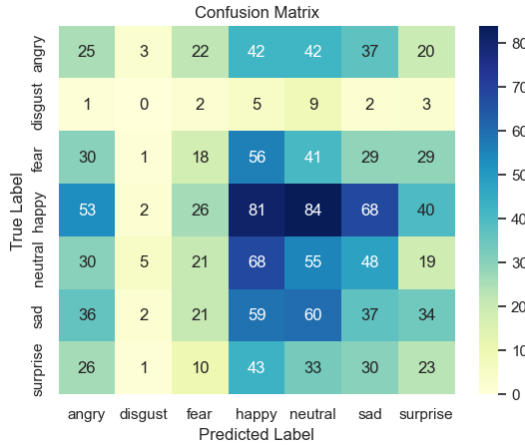


Fig. 3. Confusion matrix.

6) *Testing*: I used the `haarcascade_frontalface_default.xml` classifier from OpenCV, that is a widely used pre-trained model for detecting faces in images. Ergo, I recognized the faces with the haarcascade classifier and later recognise the facial emotion with the previous CNN. The haarcascade uses Haar features and is based on the Viola-Jones algorithm, which was one of the first methods to offer a real-time face detection solution. While it is quite efficient and works well for many applications, its performance can be limited by a complex environment, faces not in a frontal position and has lower accuracy compared to modern methods.

On the test that I did, the performance of the haarcascade was regular predicting sometimes false positives (it predicted as faces things that aren't a face) and false negatives above all when the face was not frontal. The performance of the VGG-like for facial emotion recognition wasn't good enough predicting the most of times neutral and happy when the true label were others.

B. emotion-ferplus-8

1) *Data*: I used the FER+ dataset is a notable extension of the original Facial Expression Recognition (FER) dataset (the one that I used previously). Developed to improve upon the limitations of the original dataset, FER+ offers a more refined and nuanced labeling of facial expressions. The FER+ has 8 emotions: happiness, sadness, anger, surprise, fear, disgust, neutral and contempt.

2) *RFB-320 Single Shot Multibox Detector (SSD) Model for Face Detection*: Before the emotion is recognized, the face needs to be detected in the input frame like in the previous case. For this, the Ultra-lightweight Face Detection RFB-320 is used. It is a face detection model optimized for edge computing devices. Incorporating a modified Receptive Field Block (RFB) module (an extension of traditional convolutional layers), it effectively captures multiscale contextual information without adding computational overhead. Trained on the diverse WIDER FACE dataset (32,203 images and labeled 393,703 faces with a high degree of variability in scale, pose, and occlusion) and tailored for 320×240 input resolution, it boasts an impressive balance of efficiency with a computational rate of 0.2106 GFLOPs and a compact size housing of 0.3004 million parameters. Developed in the PyTorch framework, the model achieves a commendable mean average precision (mAP) of 84.78%, marking its position as a potent solution for efficient face detection in resource-constrained environments. In this implementation, the RFB-320 SSD (object detection algorithm that detects and classifies objects in an image in a single forward pass of the network) model has been used in Caffe format (the data storage and model representation format used by Caffe, a deep learning framework). [3]

3) *Custom VGG13 Model Architecture*: The emotion recognition classification model employs a customized VGG13 architecture designed for 64×64 grayscale images. It classifies images into eight emotion classes using convolutional layers with max pooling and dropout to prevent overfitting. This enhances the model's ability to generalize and accurately recognize emotions in various images. It was training using the FER+ dataset. The structure is:

TABLE II
CUSTOMIZED VGG13 MODEL ARCHITECTURE FOR FACIAL EMOTION
RECOGNITION

Layer	Filters	Kernel Size	Activation
Input	-	$64 \times 64 \times 1$	-
Conv2D	64	3×3	ReLU
Conv2D	64	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Conv2D	128	3×3	ReLU
Conv2D	128	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Fully Connected (Dense)	1024	-	ReLU
Dropout	-	-	0.50
Fully Connected (Dense)	1024	-	ReLU
Dropout	-	-	0.50
Fully Connected (Dense)	8	-	ReLU
Softmax (Output)	8	-	Softmax

4) *Implementation*: I employed the emotion-ferplus-8.onnx model, that is a pretrained model of learnopencv, for real-time facial emotion recognition. The model was integrated with a live video stream captured via webcam. Face detection was performed using the SSD-based RFB-320 detector with a Caffe backend. Preprocessing steps included image normalization using a mean of [127, 127, 127] and standard deviation of 128.0, followed by the computation of prior boxes based on feature maps derived from multiple strides ([8, 16, 32, 64]). The detection process leveraged bounding box decoding, confidence thresholding, and Non-Maximum Suppression (NMS) with an IoU threshold of 0.3 to filter predictions. The detected face region was then passed to the ONNX model to classify into one of seven emotion categories: neutral, happiness, surprise, sadness, anger, disgust, and fear. Output video was optionally recorded for subsequent analysis.

5) *Testing*: The ultralightweight face detection RFB320 is very good and accurate recognizing the faces although the quantity, angles of the face, illumination... The custom VGG13 model architecture performed well in classifying emotions from detected faces. It excelled at recognizing happiness, neutrality, and surprise, while showing slightly lower accuracy for sadness, disgust, and fear.

IV. SPEECH EMOTION RECOGNITION

A. Data

- **Ravdess**: This portion of the RAVDESS contains 1440 files: 60 trials per actor x 24 actors = 1440. The RAVDESS contains 24 professional actors (12 female, 12 male), vocalizing two lexically-matched statements in a neutral North American accent. Speech emotions

includes calm, happy, sad, angry, fearful, surprise, and disgust expressions. Each expression is produced at two levels of emotional intensity (normal, strong), with an additional neutral expression.

- **Crema-D**: a data set of 7,442 original clips from 91 actors. These clips were from 48 male and 43 female actors between the ages of 20 and 74 coming from a variety of races and ethnicities (African America, Asian, Caucasian, Hispanic, and Unspecified). Actors spoke from a selection of 12 sentences. The sentences were presented using one of six different emotions (Anger, Disgust, Fear, Happy, Neutral, and Sad) and four different emotion levels (Low, Medium, High, and Unspecified).
- **Tess**: There are a set of 200 target words were spoken in the carrier phrase "Say the word ." by two actresses (aged 26 and 64 years) and recordings were made of the set portraying each of seven emotions (anger, disgust, fear, happiness, pleasant surprise, sadness, and neutral). There are 2800 data points (audio files) in total.
- **Savee**: The SAVEE database was recorded from four native English male speakers (identified as DC, JE, JK, KL) aged from 27 to 31 years. Emotion has been described psychologically in discrete categories: anger, disgust, fear, happiness, sadness and surprise. This is supported by the cross-cultural studies of Ekman and studies of automatic emotion recognition tended to focus on recognizing these. We added neutral to provide recordings of 7 emotion categories. The text material consisted of 15 TIMIT sentences per emotion: 3 common, 2 emotion-specific and 10 generic sentences that were different for each emotion and phonetically-balanced. The 3 common and $2 \times 6 = 12$ emotion-specific sentences were recorded as neutral to give 30 neutral sentences.

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